

UI/UX Specialist

UCS542



THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

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Assignment 5

Question1 : Write a JavaScript program to take two numbers as input from user and display their sum, difference, product, and quotient.

Terminal Output:-

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS E:\Semester 5\UI_UX\Assignments\Codes\Assignment5> node Question1.js
Enter first number: 15
Enter second number: 5
Sum: 20
Difference: 10
Product: 75
Quotient: 3
PS E:\Semester 5\UI_UX\Assignments\Codes\Assignment5> 
```

Question2: Write a JavaScript program to create an array of 5 numbers and:

- Find the largest and smallest number.
- Sort the array in ascending and descending order.

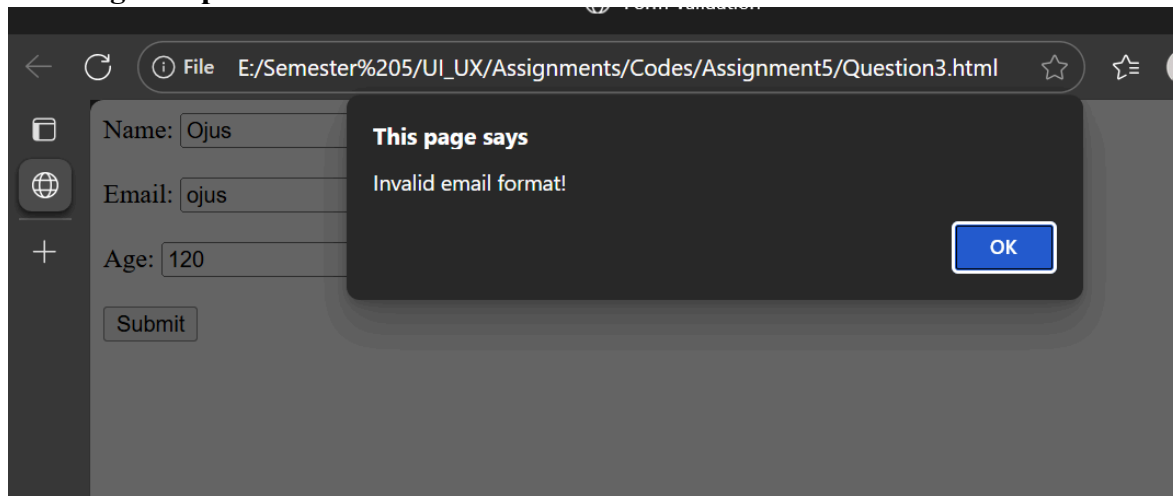
Terminal Output:-

```
PS E:\Semester 5\UI_UX\Assignments\Codes\Assignment5> node Question2.js
Array: [ 12, 5, 8, 20, 3 ]
Largest: 20
Smallest: 3
Ascending: [ 3, 5, 8, 12, 20 ]
Descending: [ 20, 12, 8, 5, 3 ]
PS E:\Semester 5\UI_UX\Assignments\Codes\Assignment5> 
```

Question3 : Create a simple form with fields for name, email, and age. Write JavaScript to validate that:

- The name field is not empty.
- The email is in the correct format.
- The age is a number between 18 and 100.

Web Page Output:-



Question4 : Write a JavaScript program to create an object representing a student (with properties like name, age, grades) and perform the following operations:

- Add a new property class.
- Update the student's grade.
- Display all the student's information.

Terminal Output:-

```
PS E:\Semester 5\UI_UX\Assignments\Codes\Assignment5> node Question4.js
name: Ojus
age: 20
grades: A+
class: 12th Grade
PS E:\Semester 5\UI_UX\Assignments\Codes\Assignment5> 
```

Question5 : Create a function that takes an array of numbers and uses the array's built-in methods (like map(), reduce(), and filter()) to:

- Remove all odd numbers.
- Multiply the remaining numbers by 2.
- Find the sum of the resulting numbers.

Terminal Output:-

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS E:\Semester 5\UI_UX\Assignments\Codes\Assignment5> node Question5.js
Even numbers: [ 2, 4, 6, 8 ]
After multiplying by 2: [ 4, 8, 12, 16 ]
Sum of resulting numbers: 40
PS E:\Semester 5\UI_UX\Assignments\Codes\Assignment5> █
```